PYTHON

Module 19: Structural Pattern Matching

Introduction to Structural Pattern Matching

Introduced in Python 3.10, the `match statement provides a powerful alternative to long chains of if-elif-else statements. At CodeHive, we refer to this as 'Structural Pattern Matching'.`

`It is cleaner, more readable, and specifically designed to handle complex data structures by matching patterns rather than just evaluating boolean logic.`

Match Syntax & Flow

The match expression is evaluated once, and its value is compared against various case patterns.

• **Evaluation**: Python checks each case from top to bottom.

• **Execution**: Once a match is found, only the code block associated with that case is executed.

• **Constraint**: Unlike some languages (like C or Java), there is no 'fall-through' in Python match statements; after a match is found, the statement completes.

The Default Case (Wildcard)

In every robust CodeHive script, it is essential to handle unexpected inputs. The underscore character _ acts as a wildcard.

• **Role**: It matches *anything* that hasn't been caught by previous cases.

• **Rule**: The case _: must always be the final entry. Placing it earlier would prevent subsequent cases from being checked.

Pattern OR Logic (The Pipe)

You can combine multiple match targets into a single block using the pipe operator `|`.

• Use Case: When different inputs should trigger the exact same logic (e.g., mapping multiple days to a single 'Weekend' category).

Example: `case 6 | 7: print("Weekend")` matches if the value is 6 OR 7.

Guard Clauses (Case Conditioning)

Guards allow you to add `if` statements directly into a `case` definition for extra precision.

- **Execution**: The case matches only if the pattern is met AND the `if` condition (the guard) evaluates to `True`.

- **Power**: This allows you to differentiate between identical values based on external state or secondary variables.

CodeHive Pro-Tips for Match

- Use `match` when you have more than 3 distinct values of a single variable to check.

- Pattern matching isn't just for numbers/strings; it can deconstruct lists and dictionaries (Advanced topic for Module 20).

- Always include a `case _:` to ensure your program doesn't silently ignore unexpected data.